Jon Morgan

Technical Director – Decarbonisation & Data Assurance

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> *"My career is built on a balance of interdisciplinary breadth—integrating defence systems research, building performance simulation, and software automation—coupled with practical experience in business operations and company leadership. I design code-governed pipelines that bridge the gap between active building systems and sustainability reporting frameworks, ensuring that automated efficiency is always structured around rigorous human oversight to translate operational metrics into verified compliance outputs. This combination of engineering depth and management experience enables me to act as a strategic advisor and practical bridge, converting complex technical datasets into the clear, audit-ready structures required by senior leadership."*

Core Capabilities

- **Building Decarbonisation & Technical Transition Modelling:** Long-term specialisation in high-fidelity simulation of energy, daylight, and building performance. Experienced in building the physical and financial models—integrating load curves, battery storage, solar sizing, and tariff arbitrage—that provide the rigorous analytical backing required for transition plans and eliminate speculative risk.
- **Building Performance & Multidisciplinary Engineering:** Ten years leading Arup's environmental and building physics team in Australasia, specialising in building performance, energy systems, and operational efficiency. Experienced in coordinating complex building parameters across services (MEP), facades, fire safety, and acoustics, alongside mentoring and guiding engineering graduates.
- **Data Assurance & Automated Pipelines:** 30 years of programming experience (Python, SQL, Bash, Linux) used to architect automated, code-governed pipelines for calculations, combining automated speed with human-in-the-loop diagnostic validation to eliminate the risk of manual errors between active plant rooms and ESG reporting platforms.
- **Strategic Operations & Governance:** Experience as a co-founder and director for nearly 8 years, managing operations, software deployments, and corporate governance. This operational background serves as a highly effective bridge, allowing me to align technical engineering delivery directly with corporate risk management, software infrastructure, and commercial property governance.

Professional Experience

Co-Perform Pty Ltd — 2018 – Present

Co-Founder & Director

Co-founded and scaled this specialized built environment consultancy, driving its technical evolution, systems automation, and computational scaling to establish a high-performance advisory framework.

- **Embodied Carbon & Building Simulation:** Managed integrated calculations for embodied carbon, daylight, and energy on a major 6-star Green Star healthcare facility. Designed and programmed a bespoke utility to parse the raw Bill of Quantities (BoQ) from the head contractor, automatically converting the data structure for direct import into the eTool carbon accounting platform.
- **Electrification & Arbitrage Modelling:** Built physical simulation models analyzing building thermal load, solar output, and battery degradation against dynamic network tariffs to evaluate the financial feasibility of behind-the-meter energy arbitrage.

- **Municipal Sustainability Policy:** Partnered with Nation Partners to co-author the City of Melbourne Sustainable Building Policy, defining standards for environmental, carbon, and operational performance across the Council's own municipal asset portfolio.
- **Planning & Building Permit Support:** Supported planning and building permit applications for commercial real estate developments by delivering high-fidelity thermal, daylight, and energy simulations that informed Sustainability Management Plans (SMPs) for municipal approvals.
- **DevOps & AI Automation:** Managed internal DevOps, constructing a custom repository that leverages AI to automate approximately 80% of ongoing deployment, testing, and cloud infrastructure tasks, ensuring core engineering oversight remains focused on strategic architecture.
- **Infrastructure Engineering:** Developed an internal cloud-based compute system using Python and Plotly Dash, allowing teams to run automated simulations securely through ngrok tunnels.
- **Systems Automation:** Automated reporting workflows by writing pipelines that sync simulation data with SQL databases (Knack) and generate PDF reports via LaTeX.
- **System Migration:** Managed web hosting transitions by rebuilding the corporate website with Astro, reconfiguring DNS, and restoring operations within 48 hours.
- **Business Strategy:** Commissioned and co-led a 6-month strategic review of business model and market alignment with Focused Solutions.
- **Operations & Governance:** Set up and managed the company's project management and CRM tools (Runn, Pneumatic, OnePageCRM, Mixmax) and led weekly project, cash flow, and directors' meetings.

Moreland Energy Foundation Ltd & Visergy

2014 – 2018

Senior Consultant (MEFL) / Principal & Sole Trader (Visergy)

Delivered high-level energy efficiency engineering, regional decarbonisation models, and performance-based design simulations for municipal portfolios and commercial real estate.

- **Sole Trader Operations (Visergy):** Established and managed an independent consultancy subcontracting to major engineering firms. Directed all commercial and operational aspects, including branding, proposal development, engineering delivery, and financial invoicing.
- **Solar Feasibility & Predictive Tooling:** Developed a fast predictive modeling tool to right-size solar systems, accelerating early-stage design feasibility and concept planning.
- **Climate-Resilient Public Housing:** Partnered with architects at DesignInc to design a template public housing prototype engineered for passive thermal resilience and occupant safety during extreme heatwaves.
- **Building Performance & Daylighting:** Provided design support for town planning and building approvals, using energy and daylight simulations to optimise performance and inform detailed Sustainability Management Plans (SMPs).
- **Regional Emissions Modelling:** Ingested and cleaned regional datasets across agriculture, residential, transport, and commercial sectors to build an emissions inventory for the Hepburn Shire/Daylesford Net-Zero transition plan.
- **Data Visualisation:** Deployed custom Python scripts to translate energy interval data into clear visual models for municipal and public stakeholders.
- **Residential Retrofit Analysis:** Modelled building physics and retrofit scenarios for public housing towers to assist the Department of Health and Human Services (DHHS) with capital upgrade decisions.

Arup

2005 – 2014

Senior Consulting Engineer — Australasian Leader in Environmental & Building Physics

Directed regional analytical capacity, workflows, hardware integration, and high-fidelity computational modelling for one of the world's premier engineering institutions.

- **Graduate Mentorship & Technical Training:** Guided and trained 3 to 4 building physics graduates over a 4 to 5-year track, and structured and delivered specialized training for MEP graduate engineers on rotation. Participated directly in graduate selection interviews.

- **Interdisciplinary Integration:** Acted as the core technical liaison on complex building projects, collaborating with teams in building services (MEP), facades, fire safety, and acoustics to align multidisciplinary design parameters with dynamic thermal models.
- **Building Performance & Quality Assurance:** Standardised computational simulation workflows across offices in Australasia, driving team rigor, auditable results, and strict quality control on dynamic compliance calculations.
- **Computation Platform & Data Assurance:** Designed and maintained an internal Linux compute cluster to centralise calculations. Standardised text-based configuration files to ensure quality control and auditable results across regional teams.
- **Software Integration:** Developed custom scripts to link Fluent CFD geometry with Radiance solar simulation, creating accurate thermal boundary conditions for complex atrium designs.

Arup (UK) & BMT Fluid Mechanics

2002 – 2005

Consulting Engineer / Analyst

Delivered advanced computational fluid dynamics (CFD), blast wave risks, and process automation across commercial real estate, sports stadia, and offshore oil/gas operations.

- **Built Environment Aerodynamics (BMT):** Utilised computational fluid dynamics (CFD) to prepare urban wind microclimate comfort and safety assessments, pollution dispersion models, and structural facade cladding wind pressure loads for complex tall buildings and precincts.
- **Wind Tunnel Experimental Validation:** Conducted light-touch experimental validation within boundary layer wind tunnels, calibrating physical model sensor data against corresponding numerical CFD simulation results to guarantee analytical assurance.
- **Process Automation:** Wrote custom Rhino3D scripts to parse coordinates from piping layouts, automatically generating 3D models for offshore explosion simulation (AutoReGas) and reducing manual modelling hours.
- **High-Performance Infrastructure:** Designed, build, and managed custom Beowulf Linux computing clusters to handle intensive queue management for heavy industrial hydrodynamic and structural stress simulations.

Defence Science and Technology Organisation (DSTO)

1996 – 2000

Research & Development Systems Engineer

Executed advanced research into aerodynamic, hydrodynamic, and gas turbine performance metrics for defence assets and weapons systems.

- **Gas Turbine Fault Detection:** Developed an optimisation pipeline for Pratt & Whitney F404 engines. Wrapped a legacy Fortran 90 engine model in MATLAB and wrote a Nelder-Mead (Downhill Simplex) algorithm to match physical test-bed logs with simulation outputs to diagnose engine degradation.

Education & Accreditations

Professional Credentials & Industry Accreditations

- **Accredited Assessor — NABERS Upfront Carbon** (Embodied carbon assessment for materials, structure, and supply-chain metrics).
- **Energy & Sustainability Hub Advisory Board Member** — Building Designers Association of Victoria (2015 – 2018).

RMIT University

2009 – 2014

Master of Urban Planning and Environment

Advanced postgraduate studies focusing on urban environmental policy, microclimate resilience, sustainable infrastructure, and quantitative resource modelling.

RMIT University

1991 – 1995

Bachelor of Engineering (B.Eng.), Aerospace

Rigorous four-year engineering specialisation with deep focus on fluid dynamics and aerodynamics, structural mechanics and aerostructures, aircraft systems, dynamic systems and control, thermodynamics, and computational simulation.

Compliance & Operations

My approach relies on a strategic balance: the broad, cross-disciplinary engineering background required to trace data across a portfolio's entire physical footprint, alongside a deep focus on the exact touchpoints where active building operations meet corporate compliance structures. Rather than prescribing a one-size-fits-all framework, I collaborate to map out precise technical connections—bridging utility metering setups, HVAC performance, and embodied carbon data directly into your corporate reporting systems.

For institutional property portfolios, navigating decarbonisation requires converting transition plans from high-level estimates into defensible capital investments. My depth in physical simulation—incorporating localized load curves, battery storage parameters, and tariff arbitrage models—allows me to deliver highly validated, adaptive models. By grounding transition strategy in verified, data-backed models, I help management teams confidently align technical plans with broader financial and corporate investment goals.

With a career that spans both technical engineering and business leadership, I view compliance and decarbonisation as integrated, operational challenges. I bring a rare toolkit: the technical depth to understand complex systems, paired with the commercial literacy to manage projects, budgets, and risk. I look forward to partnering with your teams to configure the automated, verifiable pipelines—grounded in experienced human-in-the-loop validation—that remove tracking risks, protect asset performance, and support absolute reporting integrity on the corporate balance sheet.

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